

Standard PyroSim Software Training Template

Fire & Smoke Modeling with PyroSim (FDS Graphical Interface)

COURSE OVERVIEW

PyroSim is a graphical user interface for the Fire Dynamics Simulator (FDS), developed by the National Institute of Standards and Technology (NIST). This training equips participants with practical skills to build, run, and analyze computational fire and smoke simulations for building and life-safety applications.

LEARNING OBJECTIVES

- Navigate the PyroSim interface and project setup workflow
- Build accurate 3D geometry: compartments, obstructions, vents
- Define materials, surfaces, and reaction (fuel) properties
- Place fire sources, HVAC, sprinklers, and detection devices
- Configure mesh resolution and run FDS simulations
- Visualize and interpret results (Smokeview, slices, devices)
- Validate results against design objectives / code criteria

WHO SHOULD ATTEND

- Fire protection engineers
- Life safety / egress consultants
- Building & code compliance professionals
- Mechanical / HVAC smoke-control designers
- AHJ reviewers evaluating performance-based designs

Prerequisites

- Basic fire dynamics knowledge recommended
- Laptop with PyroSim installed (trial license provided)

TRAINING AGENDA (1-DAY, 6 HOURS)

Time	Topic
9:00 – 9:30	Introduction to fire modeling, FDS, and the PyroSim workflow
9:30 – 10:30	Creating geometry: compartments, meshes, obstructions & vents
10:30 – 11:15	Materials, surfaces, and defining the fire (reaction & HRR)
11:15 – 12:00	Devices, detectors, sprinklers & HVAC / smoke control systems
12:00 – 12:45	Lunch break
12:45 – 1:45	Running simulations & mesh/time-step best practices
1:45 – 2:45	Post-processing: Smokeview, slices, isosurfaces, plots
2:45 – 3:30	Hands-on case study: performance-based design scenario
3:30 – 4:00	Q&A, resources, and next steps

Deliverables: sample project files, certificate of completion (Optional).

Format: In-person or virtual, max 12 participants.